

Introduction

To



COSC 205

Objectives

At the end of this class, student should be able to understand;

- 1. The history of Java**
- 2. The idea of Java Class Libraries**
- 3. The basics of a typical Java Environment and its phases.**
- 4. The characteristics of Java**
- 5. How to run a simple Java code**

History of Java

- Java

- Based on C and C++
- Developed in 1991 for intelligent consumer electronic devices
 - Market did not develop, project in danger of being cancelled
- Internet exploded in 1993, saved project
 - Used Java to create web pages with dynamic content
- Java formally announced in 1995
- Now used to create web pages with interactive content, enhance web servers, applications for consumer devices (pagers, cell phones)...

What does Java mean?

Java is a high-level programming language developed by Sun Microsystems. **Java** programs are interpreted by the **Java** Virtual Machine, or JVM, which runs on multiple platforms. This means all **Java** programs are multiplatform and can run on different platforms, including Macintosh, Windows, and Unix computers.

What is the full form of Java?

"**Java**" being used as slang for "coffee"

Java does not have any **full form**, but a programming language originally developed by James Gosling at Sun Microsystems in 1995. It derives much of its syntax from the most popular programming languages of all time : C and C++.

Why is it called Java?

The language was initially **called** Oak after an oak tree that stood outside Gosling's office. Later the project went by the name Green and was finally renamed **Java**, from **Java** coffee. Gosling designed **Java** with a C/C++-style syntax that system and application programmers would find familiar.

Java Class Libraries

- Java programs
 - Consist of pieces called classes
 - Classes contain methods, which perform tasks
- Class libraries
 - Also known as Java API (Applications Programming Interface)
 - Rich collection of predefined classes, which you can use
- Two parts to learning Java
 - Learning the language itself, so you can create your own classes
 - Learning how to use the existing classes in the libraries

Basics of a Typical Java Environment

Java Systems

- Consist of environment,
- language,
- Java Applications Programming Interface (API),
- class libraries

5 Phases of Java Programs

1. Edit

- Programmer writes program (and stores program on disk)

2. Compile

- Compiler creates *bytecodes* from program

3. Load

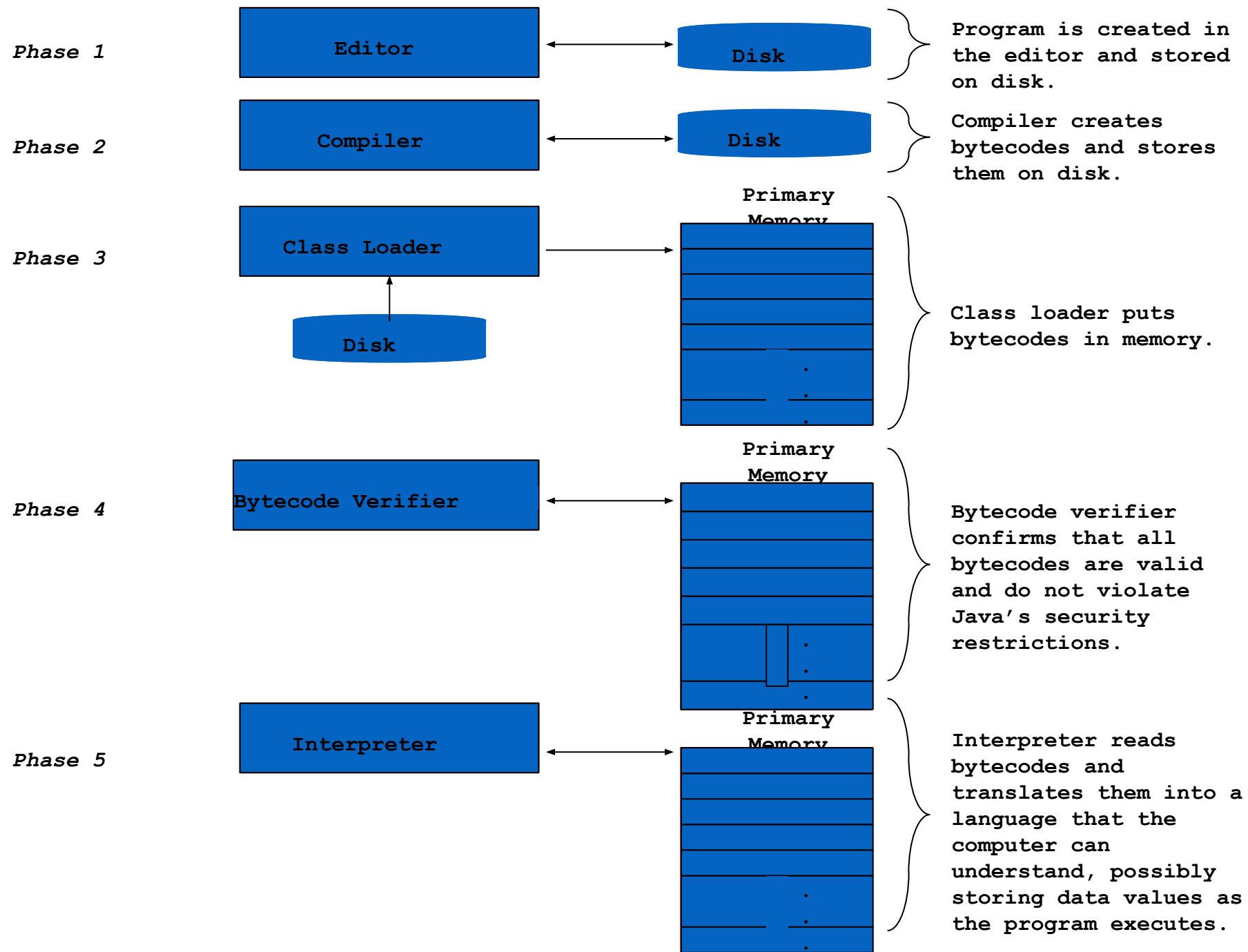
- Class loader stores bytecodes in memory

4. Verify

- Verifier ensures bytecodes do not violate security requirements

5. Execute

- Interpreter translates bytecodes into machine language



Features/Characteristics of Java

1. Simple :

- Java is Easy to write and more readable and eye catching.
- Java has a concise, cohesive set of features that makes it easy to learn and use.
- Most of the concepts are drew from C++ thus making Java learning simpler.

2. Secure :

- Java program cannot harm other system thus making it secure.
- Java provides a secure means of creating Internet applications.
- Java provides secure way to access web applications.

Features/Characteristics of Java

3. Portable :

- Java programs can execute in any environment for which there is a Java run-time system.(JVM)
- Java programs can be run on any platform (Linux,Window,Mac)
- Java programs can be transferred over world wide web (e.g applets)

4. Object-oriented :

- Java programming is object-oriented programming language.

Features/Characteristics of Java

5. Robust :

- Java encourages error-free programming by being strictly typed and performing run-time checks.

6. Multithreaded :

- Java provides integrated support for multithreaded programming.

7. Architecture-neutral :

- Java is not tied to a specific machine or operating system architecture.
- Machine Independent i.e Java is independent of hardware .

Features/Characteristics of Java

8. **Interpreted :**

- The Bytecode can be interpreted on any platform by JVM.

9. **High performance :**

- Bytecodes are highly optimized.
- JVM can execute them much faster .

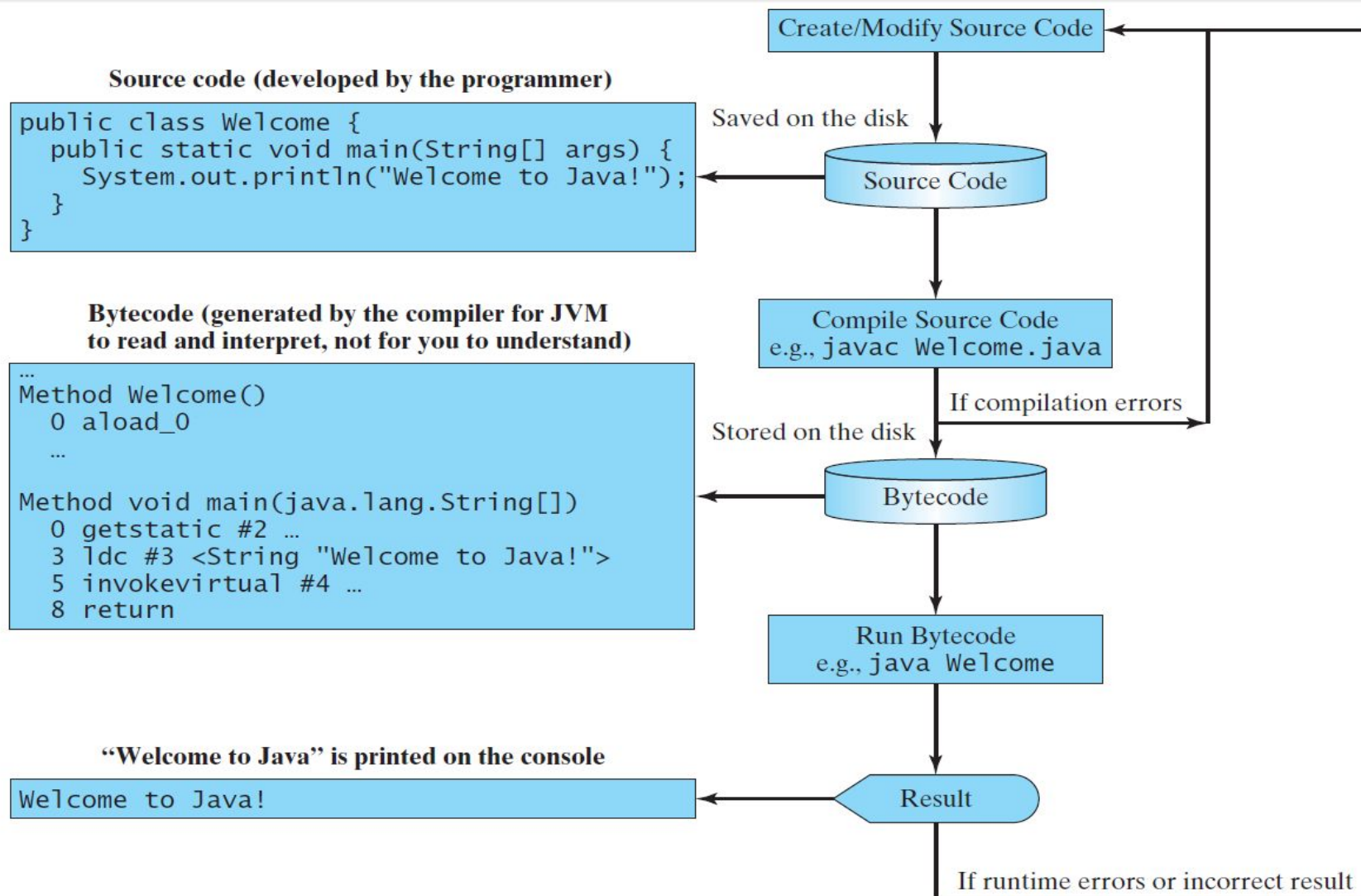
10. **Distributed :**

- Java was designed with the distributed environment.
- Java can be transmit,run over internet.

11. **Dynamic :**

- Java programs carry with them substantial amounts of run-time type information that is used to verify and resolve accesses to objects at run time.

How to run a simple Java code



Class Participation

You may write an interactive java program or a hard coded program that reads a Celsius degree in double from the console, then converts it to Fahrenheit and displays the result. The formula for the conversion is as follows:

$$\text{fahrenheit} = (9 / 5) * \text{celsius} + 32$$

Hint: In Java, $9 / 5$ is 1 , but $9.0 / 5$ is 1.8 .